MEDICAL CORE · INTEGRATED SOFTWARE

Surgical Navigation Systems

Registration results become a spatial experience users can understand and trust.

EVIDENCE STATE Ongoing	**PERIOD** 2023.07 - present
PORTFOLIO TIER Medical Core	**PRIMARY CAPABILITY** 3D Geometry & Registration

Connect tracking devices, SDKs, coordinate transforms, data flow, and HoloLens spatial presentation into one surgical-navigation experience.

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Problem and system boundary

01 / COORDINATE CHAIN

Coordinate chain

Made the transform path visible from tracking hardware through images, tools, and HoloLens.

PROBLEM

When device coordinates and registration state stay fragmented, users cannot confidently interpret the spatial result.

Connect tracking devices, SDKs, coordinate transforms, data flow, and HoloLens spatial presentation into one surgical-navigation experience.

PUBLIC BOUNDARY

The demonstration shows integrated operation on a phantom; it does not claim clinical efficacy or production deployment.

Joint review spans tracking hardware, medical imaging, XR, and workflow expertise.

EVIDENCE STATE

Ongoing / Demonstration clip from the HoloLens viewpoint: a hologram registered on a skull phantom with a tracked pointer.

Owned decisions and implementation

MY ROLE

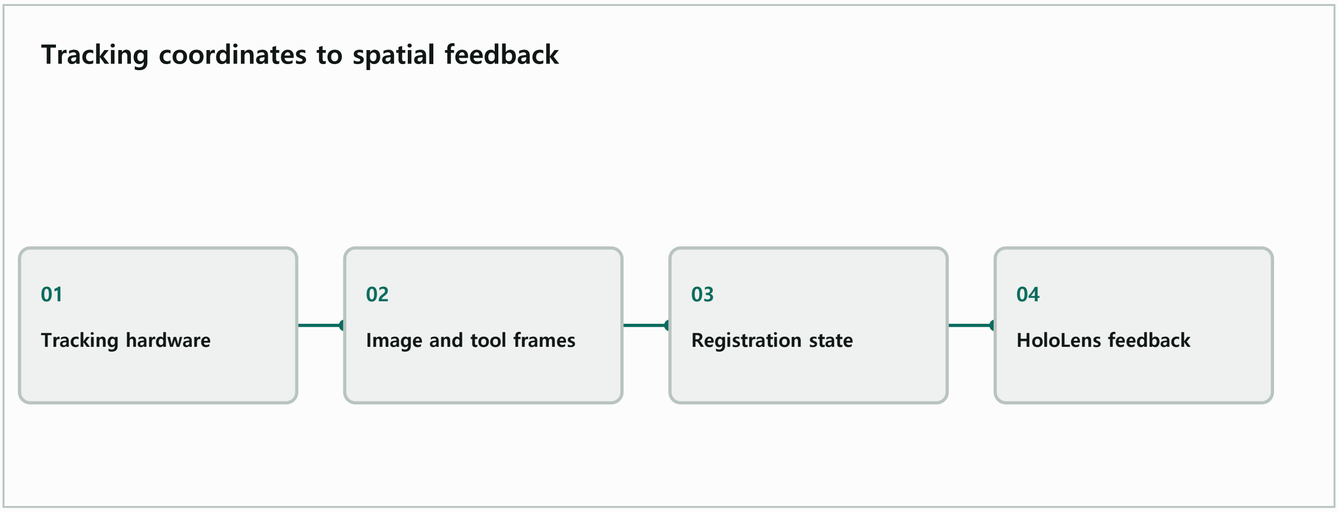
Led the integrated software across device, SDK, coordinate transforms, and data flow, plus the HoloLens spatial-placement and registration-feedback experience.

02 / SPATIAL FEEDBACK

Spatial feedback

Made registration state legible through placement and feedback rather than a hidden number.

SYSTEM RELATIONSHIP



Explanatory system relationship diagram; it is not photographic or experimental evidence.

IMPLEMENTATION TECHNOLOGIES

HoloLens 2 / Optical tracking / 3D Slicer / Unity / MRTK / OpenIGTLink

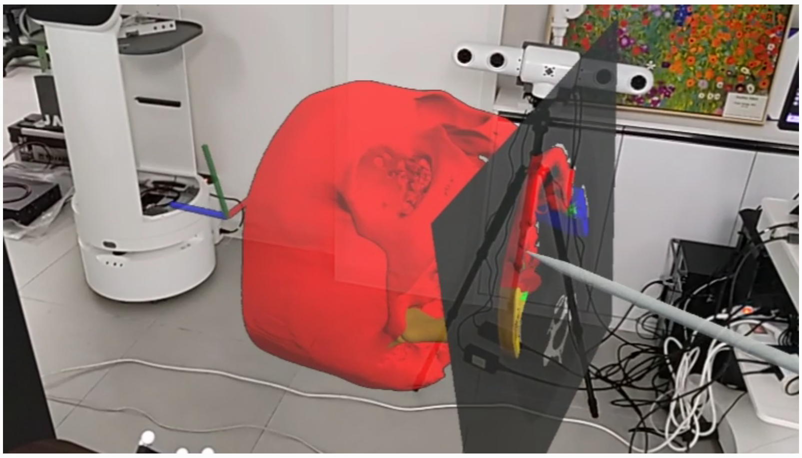
Verification and evidence ledger

03 / DEMONSTRATION EVIDENCE

Demonstration evidence

Treat the working integration demo as evidence while keeping media approval explicit.

IMAGE / PUBLICLY INSPECTABLE



REGISTERED PUBLIC EVIDENCE

VIDEO / PUBLICLY INSPECTABLE

surgical-navigation-clip-01

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

surgical-navigation-poster-01

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

surgical-navigation-gallery-01

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

surgical-navigation-gallery-02

Approved derivative; caption in portfolio data.

IMAGE / PUBLICLY INSPECTABLE

surgical-navigation-gallery-03

Approved derivative; caption in portfolio data.

Role, team result, and limits

04 / CLINICAL BOUNDARY

Clinical boundary

An integration demonstration does not establish clinical efficacy or production deployment.

MY ROLE

Led integrated software and HoloLens spatial placement and registration feedback.

Led the integrated software across device, SDK, coordinate transforms, and data flow, plus the HoloLens spatial-placement and registration-feedback experience.

TEAM RESULT

The team and external collaborators jointly conducted integration demonstrations and acceptance reviews; those results are not attributed

Working device connections, coordinate transforms, spatial placement, and registration-feedback demonstrations provide the evidence.

LIMITATIONS

The demonstration shows integrated operation on a phantom; it does not claim clinical efficacy or production deployment.

Demonstration clip from the HoloLens viewpoint: a hologram registered on a skull phantom with a tracked pointer.

COLLABORATION BOUNDARY

Joint review spans tracking hardware, medical imaging, XR, and workflow expertise.

Connect tracking devices, SDKs, coordinate transforms, data flow, and HoloLens spatial presentation into one surgical-navigation experience.

Recap and joint-development invitation

Registration results become a spatial experience users can understand and trust.

PRIMARY CAPABILITY

3D Geometry & Registration / Medical Navigation & Visualization / XR Application Engineering

REGISTERED PUBLIC EVIDENCE

No external public link is currently registered.

WHAT TO BRING

Share the problem, data or sensors, target validation, and schedule. We can define coordinate and evidence boundaries together.

Joint review spans tracking hardware, medical imaging, XR, and workflow expertise.

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